

Model Refinement and Test Submission

Retail Location Intelligence | AI-supported market planning in Çankaya

Purpose: help café entrepreneurs shortlist candidate areas faster by comparing access to their intended customer surroundings and competition. Access uses indirect spatial indicators; time savings remain an unmeasured pilot goal.

Group 6 · Technical baseline: 58ce550 · Narrative updated 6 October 2026

1. Overview and initial evaluation

Refinement focused on evaluation validity rather than searching for a higher score. The earlier implementation allowed café presence into POI diversity, used full-data median imputation before splitting, and blended held-out and training-data predictions. Those results cannot validate the corrected model. No controlled before/after performance comparison is claimed.

2. Refinement, feature selection and tuning

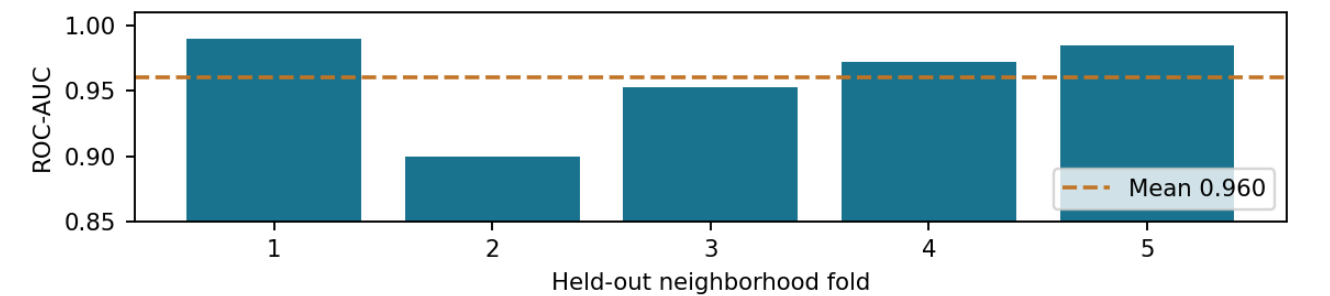
Café-free diversity is rebuilt from count columns, including for the legacy Parquet file. Café counts are excluded from the 12 RF predictors; they define only the target, $\text{cafes}_{500\text{m}} \geq 1$. Each fold fits medians on training cells; all-missing training columns fall back to zero. The displayed score is $100 \times$ held-out probability, without a full-data prediction blend. Missing target values, a single target class or too few groups stop training.

Fixed configuration: 300 trees, $\text{max_depth}=8$, $\text{min_samples_leaf}=5$, $\text{class_weight}=\text{"balanced"}$, $\text{random_state}=42$ and $\text{n_jobs}=-1$. No GridSearchCV, randomized search, recursive feature elimination or alternative-model benchmark is implemented. Feature changes remove leakage; they are not an optimized subset search [1, 2].

3. Cross-validation and final evaluation

StratifiedGroupKFold uses five shuffled folds, seed 42, grouped by `mahalle_name`. A neighborhood never appears in training and assessment for the same fold. Neighboring areas across boundaries may still share buffers; exclusion-buffer validation remains future work [3].

Fold	Train cells	Held-out cells	Positive rate	ROC-AUC
1	4,262	1,099	14.1%	0.990
2	4,318	1,043	17.1%	0.900
3	4,306	1,055	15.9%	0.953
4	4,275	1,086	14.2%	0.972
5	4,283	1,078	15.0%	0.985



Mean fold ROC-AUC: 0.960. This is the mean of the five reported rounded fold values. Run: 5 October 2026; Python 3.12.14, scikit-learn 1.8.0, GeoPandas 1.2.0. The 5,361-cell snapshot was not refreshed. Fold variability (0.900-0.990) motivates stronger spatial evaluation [1].

4. Test preparation and model application

There is no separately reserved external test dataset or competition submission. Assessment uses held-out neighborhoods within cross-validation. For each fold, numeric predictors and café-free diversity are prepared, the training median is applied to both partitions, the RF is fitted on training cells, and predictions are placed back in the held-out cells' original positions. No café-presence labels are used as RF predictors.

```
import geopandas as gpd
from src.cafe_similarity import train_and_score

grid = gpd.read_parquet(
    "data/processed/cankaya_grid_features.parquet")
scored, report = train_and_score(grid)
print(report[["fold", "roc_auc"]])
# Computes held-out scores; does not write the dataset.
```

5. Test metrics and implementation checks

ROC-AUC evaluates discrimination for mapped café presence. Accuracy, F1, calibration error and a threshold-based confusion matrix are not reported in the recorded validation; no decision threshold was validated. Training-set metrics and independent test metrics are unavailable. An ROC curve requires raw predictions; the fold-AUC chart summarizes the existing validation without inventing a curve.

On the merged main commit, all 43 tests passed under Python 3.11 and 3.12 in GitHub Actions. Tests cover scoring, feature leakage, fold-specific imputation, held-out score construction, competition thresholds, school eligibility and Streamlit flows. Software correctness checks are separate from model validation and the business KPI.

6. Deployment and interpretation

The Streamlit app recomputes similarity at runtime and caches the resulting data. An additional all-data RF fit supplies Gini feature importances only; it does not replace held-out displayed scores. The model is not saved as a serialized inference artifact. This is a grid-comparison prototype rather than an independently tested commercial prediction service.

7. Conclusion and remaining work

Refinement removes direct target leakage and prevents training predictions from inflating the displayed score. The strongest reported Gini inputs are café-free diversity (23.9%), restaurants (16.9%) and shops (16.3%). These are associations, not causal drivers. Next steps are a separate geographically buffered evaluation, calibration checks, a simple baseline and the proposed timed shortlist pilot. Revenue or profitability claims require new labels.

References and implementation evidence

[1] Group 6, implementation baseline (src/cafe_similarity.py, docs/model_validation.md, tests, app.py); reviewed 6 October 2026. https://github.com/edasaruhan/SIC_AI_17_Capstone_Group_6/tree/58ce550c027d3ddce9f209fe3979bea849d5716a

[2] scikit-learn, RandomForestClassifier documentation. <https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestClassifier.html>

[3] scikit-learn, StratifiedGroupKFold documentation. https://scikit-learn.org/stable/modules/generated/sklearn.model_selection.StratifiedGroupKFold.html